GAP-10T Split-Jet Auxiliary Completion: Nonpropagation, On-Shell Decoupling, and the Remaining Selection No-Go

Ing. David Jaroš

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Abstract

The split-jet right inverse represents every local Lorentz tetrad while the physical connection remains Levi–Civita, but the previous construction was kinematic. We give an action-level auxiliary implementation and prove its complete local Euler–Lagrange structure. A Lagrange-multiplier constraint forces the split-jet equation. Variation of the Lorentz jet tensor and the relative-central one-form then forces the multiplier itself to vanish on every non-null patch. Consequently the jet variables carry no derivatives, no propagating modes, no stress tensor, and no spin source on shell. When this sector is added to any gravitational effective action, the metric equation is unchanged while every tetrad remains locally representable by the single biquaternionic jet. We also prove the exact limitation: because the algebraic jet map is surjective, this constraint sector cannot select a tetrad from Θ. A separate metric effective action or a genuinely new dynamical relation is logically necessary.

1 Fields and constraint

Let ${e_\mu}^a$ be a nondegenerate tetrad, let $\mathring{\omega}(e)$ be its Levi–Civita spin connection, and let $X = X^a \mathbf{u}_a$ be the Lorentz-real projection of Θ on a patch where

$$X^2 := \eta_{ab} X^a X^b \neq 0, \qquad s := \sqrt{\mathcal{N}_0}. \tag{1}$$

Introduce a Lorentz-algebra-valued jet variable $K^J_{\mu ab} = -K^J_{\mu ba}$, a real relative-central one-form w_μ, and a vector-valued Lagrange multiplier ${\lambda_a}^\mu$. None of these objects is used in physical curvature. Define

$${C_\mu}^a := \mathring{D}_\mu X^a + {K^J}^a{}_{\mu b} X^b + w_\mu X^a - s{e_\mu}^a \tag{2}$$

and the auxiliary action

$$\boxed{S_{\text{aux}}[e, X, K^J, w, \lambda] = \int d^4x\, \sqrt{-g}\; {\lambda_a}^\mu {C_\mu}^a.} \tag{3}$$

This is a constraint sector, not an additional kinetic sector.

2 Euler–Lagrange theorem

Theorem 1 (Auxiliary split-jet closure)**.** *On every patch with $X^2 \neq 0$, the Euler–Lagrange equations of Eq.* (3) *imply*

$${C_\mu}^a = 0, \qquad {\lambda_a}^\mu = 0. \tag{4}$$

Thus K^J, w, and λ add no local propagating degree of freedom and the auxiliary sector contributes neither a tetrad source nor a physical spin current on shell.

Proof. Variation with respect to ${\lambda_a}^\mu$ gives ${C_\mu}^a = 0$. Variation with respect to w_μ gives

$$ {\lambda_a}^\mu X^a = 0. \tag{5} $$

Because $K^J_{\mu ab}$ is antisymmetric, its variation gives

$$ \lambda^{\mu[a} X^{b]} = 0. \tag{6} $$

For fixed μ, Eq. (6) says that λ^μ and X are linearly dependent: ${\lambda_a}^\mu = \alpha^\mu X_a$. Substitution into Eq. (5) gives $\alpha^\mu X^2 = 0$. Since $X^2 \neq 0$, $\alpha^\mu = 0$, proving Eq. (4).

The variables K^J, w, and λ occur without derivatives. Their field equations are therefore algebraic. Variations of e, X, and the physical Levi–Civita connection arising from S_{aux} are proportional to λ or its covariant derivative after integration by parts. They vanish on the solution $\lambda = 0$. Hence the constraint sector has no on-shell stress or spin source and no propagator of its own. □

3 Right inverse and stabilizer

The constraint is solvable for every (e, X). Put

$$ {Z_\mu}^a := s{e_\mu}^a - \mathring{D}_\mu X^a, \qquad w_\mu := \frac{X_a {Z_\mu}^a}{X^2}, \qquad {Z_{\mu\perp}}^a := {Z_\mu}^a - w_\mu X^a. \tag{7} $$

Then the representative

$$ K^J_{\mu ab} = \frac{Z_{\mu\perp a} X_b - X_a Z_{\mu\perp b}}{X^2} \tag{8} $$

satisfies ${K^J}^a{}_{\mu b} X^b = {Z_{\mu\perp}}^a$ and hence $C = 0$. The general solution contains an algebraic stabilizer $C^{\text{stab}}_{\mu ab} X^b = 0$. Setting this component to zero is a local representative choice; it is invisible in the tetrad and physical curvature.

Corollary 2 (No backreaction on a gravitational effective action)**.** *Let $\Gamma_{\text{grav}}[e]$ be any diffeomorphism- and Lorentz-invariant functional of the physical tetrad and its Levi–Civita curvature. The combined action*

$$ \Gamma_{\text{tot}} = \Gamma_{\text{grav}}[e] + S_{\text{aux}} \tag{9} $$

has exactly the metric/tetrad equation of Γ_{grav} on shell, while $C = 0$ supplies a local single-Θ jet representation of every solution.

4 Selection no-go for the pure constraint branch

Proposition 3 (Surjectivity prevents tetrad selection)**.** *For every non-null X and every tetrad e, there exist algebraic (K^J, w) solving $C = 0$. Therefore the pure auxiliary action Eq. (3) imposes no equation selecting one tetrad e from X or Θ. Eliminating the auxiliary variables cannot produce a nontrivial metric potential: the constraint is solvable with zero action for every tetrad.*

Proof. Equation (8) is an explicit right inverse for arbitrary $Z = se - \mathring{D}X$. On $C = 0$, $S_{\text{aux}} = 0$, and Theorem 1 shows that its first variation with respect to physical variables vanishes as well. Hence no tetrad is preferred by this sector. □

This is not a defect of the algebraic right inverse; it is the precise price of universal representability. The physical tetrad must be selected by a separate effective action, or the fundamental theory must supply an additional non-surjective dynamical relation.

Exact status

GAP-10T-JET-AUX: CLOSED [L1]. The split-jet relation has an explicit local action implementation. Its jet variables are algebraic, nonpropagating, and decouple from the physical field equations on shell.

GAP-10T-JET-CONSTRAINT-SELECTION: CLOSED AS NO-GO [L1]. A pure multiplier implementation of the surjective right inverse cannot select $E[\Theta]$; every tetrad is admissible.

GAP-10T-JET-DYN: NARROWED. Nonpropagation and on-shell decoupling are closed. What remains is the origin of the separate metric effective action (or another non-surjective dynamical principle), null-patch/global continuation, and a complete quantum degree-of- freedom measure.